When is a passage time exactly computable? The noisy saddle-node bifurcation

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Abstract

Near a saddle-node bifurcation, a noisy one-dimensional system reduces after rescaling to a single equation with quadratic drift, the normal form of the bifurcation; in neuroscience, the quadratic integrate-and-fire neuron. A particle entering from the far left escapes to the right after a random time: when can such a time be computed exactly? The mean is a double integral, and a cubic primitive is exactly what makes its inner integration Gaussian; among polynomial drifts this singles out the quadratic one, so the universal normal form and the case this route solves coincide. For any drift, the generator of the diffusion is conjugate to a supersymmetric Schrödinger operator whose superpotential is the drift, so supersymmetry is a property of the diffusion, not an assumption; under boundary hypotheses a classical expansion then gives the passage time as a sum of independent exponential variables with the eigenvalues as rates, so computing the law becomes computing a spectrum. For the normal form the mean is a product of Airy functions, for every complex value of the parameter; the tail decays at the Kramers rate, with its prefactor; and an unusually quick passage has probability decaying like the exponential of an explicit constant over the cube of the time: both ends and the bulk. Its Laplace transform satisfies an exact cubic functional equation, placing it on the monodromy surface of the second Painlevé equation.

Keywords: saddle-node bifurcation, passage time, spectral determinant, supersymmetric quantum mechanics, quartic oscillator, Airy functions, spectral zeta function, Kramers rate, quadratic integrate-and-fire neuron, second Painlevé equation, functional equation

MSC Classification: 60J60 , 34L40 , 81Q60 , 33C10 , 34M55 , 92C20

1 Introduction

A one-dimensional system whose equilibrium disappears in a saddle-node bifurcation is, near the bifurcation and after a smooth change of variable and a rescaling, described by a quadratic drift: this is the normal form of the bifurcation [36]. Adding noise of fixed intensity gives one equation, and not a family of them, which is why that equation deserves to be studied on its own. It occurs in neuroscience as the quadratic integrate-and-fire neuron, the onset of firing of a type I neuron being a saddle-node bifurcation [30]. A particle entering from the far left drifts, is shaken by the noise, lingers near the vanished equilibrium, and finally escapes to the right. How long does this take?

The mean of that time is known in several forms [5, 6, 30], and in a closed form in Airy functions, for real values of the bifurcation parameter, by Hathcock and Sethna [24, Eq. (15)]. The distribution itself has been described through the eigenvalues of the Fokker–Planck operator with an equal-spacing approximation [24], and a general eigenvalue expansion for hitting times of diffusions is due to Kent [29], the discrete counterpart being a theorem of Keilson for birth-and-death chains [19, 28]. What is missing is the exact law, and an account of why this particular drift should be the one for which exact answers exist.

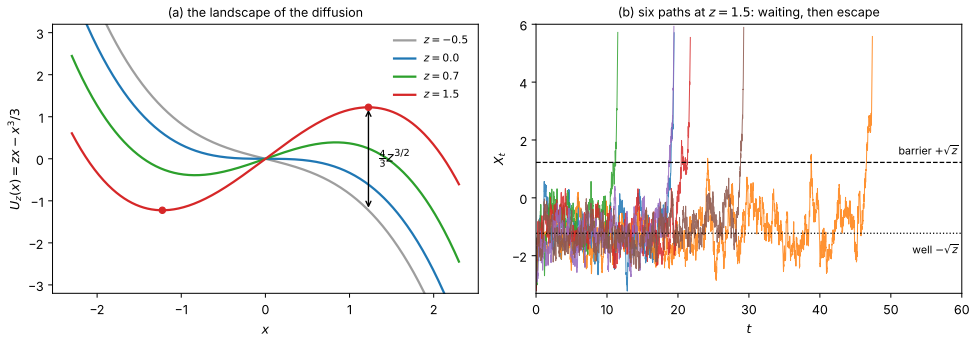


Fig. 1 (a) The landscape of the diffusion, U_z with $U_z' = -f_z$, for four values of the bifurcation parameter. For $z > 0$ it has a well at $-\sqrt{z}$ and a barrier at $+\sqrt{z}$, whose height is $\frac{4}{3}z^{3/2}$; this is the exponent of the Kramers rate of Corollary 21. For $z \le 0$ there is no barrier and the particle slides through. (b) Six paths at $z = 1.5$, started at $x = -3$ with step $2 \cdot 10^{-4}$: each fluctuates about the well, reaches above the barrier several times without escaping, and finally leaves. The passage time is the time of that escape.

Figure 1 shows the landscape and a few paths. This paper supplies the law and that account, and the answer separates into three levels, which we state in the order in which they are proved.

Any drift. Let f be a drift, F one of its primitives, and consider the diffusion with generator $L = \partial_x^2 + f\partial_x$. Conjugating by $\mathrm{e}^{F/2}$ turns $-L$ into the Schrödinger operator $H = -\mathrm{d}^2/\mathrm{d}x^2 + \frac{1}{4}f^2 + \frac{1}{2}f'$, which factors as $A^\dagger A$ with $A = \partial_x - \frac{1}{2}f$. So H is supersymmetric, with superpotential $f/2$, for every drift: supersymmetry here is a property

of the diffusion, not an additional assumption, and this is the reason an explicit zero-energy solution is available throughout. Under hypotheses on the two boundaries, the passage time is distributed as $\sum_k E_k/\theta_k$, where the θ_k are the eigenvalues of H and the E_k are independent exponential variables of mean 1. Expansions of this kind go back to Kent [29], whose Corollary 5.1 is the case of a target interior to the state space; Proposition 5 is the case where the target is the boundary itself, reached by explosion. What the supersymmetric reading adds is not the expansion but the operator, which is explicit for every drift and which the following sections can therefore compute with. Its cumulants are then the values of the spectral zeta function of H at the positive integers (Corollary 10), and the coefficients of the spectral determinant of H are ordered integrals of $\mathrm{e}^{\pm F}$ (Proposition 6). The normalisation which makes this work is that L annihilates the constants, that is, that the diffusion has no killing.

The quadratic drift. For $f_z(x) = x^2 - z$, the normal form with bifurcation parameter z, the operator H_z is a supersymmetric quartic oscillator, and the mean passage time has a closed form (Theorem 16): it equals $\pi^2(\mathrm{Ai}(z)^2+\mathrm{Bi}(z)^2)$, where Ai and Bi are the Airy functions [31, Ch. 9], for every complex z, and in particular in the double-well regime $z > 0$, where the escape is exponentially slow. The exponential rate of the tail of the law is then the smallest eigenvalue, for which we obtain $\theta_1(z) \sim \pi^{-1}\sqrt{z}\,\mathrm{e}^{-4z^{3/2}/3}$ as $z \to +\infty$ (Corollary 21), the Kramers rate with its prefactor, whose exponent is twice the instanton action of supersymmetric quantum mechanics [43]. At the opposite end, the behaviour of the determinant for large values of the spectral variable gives the probability of an unusually quick passage: $-\log \mathbb{P}(\tau \le t)$ is equivalent to $\Gamma(1/4)^8/(192\pi^2 t^3)$ as $t \downarrow 0$ (Proposition 25), a constant which does not depend on z. The law of the passage time is thus described at both ends and in the bulk.

Why this drift and no other. The closed form is not an accident of the computation, and it does not survive a change of drift. The mean is a double integral of $\mathrm{e}^{F(x)-F(x')}$ over $x < x'$, and the substitution $x = y - \ell/2$, $x' = y + \ell/2$ makes the exponent $y^2\ell + \ell^3/12 - z\ell$, which is quadratic in y precisely because F is cubic; the integration in y is then Gaussian and what remains is an Airy integral. For a polynomial drift of any other degree the exponent is not quadratic in y, the Gaussian collapse does not occur, and no such closed form is produced by this mechanism (Proposition 18). The quadratic drift is thus both the universal normal form and the only polynomial drift for which the method closes, which is the reason the two coincide in the literature.

A fourth fact, of a different nature, concerns the determinant itself. For the normal form, and in fact for a quartic oscillator with an arbitrary linear term, the determinant satisfies a cubic functional equation (Theorems 32 and 33), which places the triple of its rotated values on the monodromy surface of the second Painlevé equation. The Laplace transform of the passage time therefore satisfies an exact algebraic relation, which we prove and use, and which we place in its context in Section 9.

Organisation

Section 2 treats an arbitrary drift. Section 3 specialises to the normal form, Section 4 proves the exact law, Section 5 the closed form of the mean and its uniqueness, Section 6 the cumulants, Section 7 the tail and Section 8 the small times, which closes the description of the law. Section 9 then proves the functional equation and places

it on the monodromy surface; its first subsection is technical and may be skipped. Section 10 concludes. A companion paper [27] studies the quartic oscillator with a general linear term for its own sake, and proves there several of the statements recalled in Sections 8 and 9, among them Proposition 24, which the present paper uses only in Remark 19 and Corollary 41. Numerical checks are collected in Appendix A and the notation in Appendix B.

2 Any drift: the spectral law of the passage time

We begin with an arbitrary drift, so that the role played later by the quadratic term is visible. Nothing in this section uses the shape of the drift beyond the hypotheses stated, and the normal form only enters in Section 3.

Definition 1 Let $f \in C^1(\mathbb{R})$, let $F(x) = \int_0^x f$, and let $L = \partial_x^2 + f\partial_x$. Let W be a standard Brownian motion. For $x \in \mathbb{R}$, let X^x be the solution of $\mathrm{d}X_t = f(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t$ started at x, and let τ_x be its explosion time. Write $\mathrm{Wr}(u, v) = uv' - u'v$.

The operator L is the generator of this diffusion, and the factor $\sqrt{2}$ normalises the noise; any other intensity is recovered by rescaling [30, Sec. 2]. Lemma 2 rewrites L as a Schrödinger operator. The conjugation is classical, and so is the resulting factorisation, which is the ground-state transformation of supersymmetric quantum mechanics [7, 43]. What matters here is not the identity but its scope: it holds whatever the drift, so that an explicit zero-energy solution is available throughout, and this is the mechanism behind every closed form below.

Lemma 2 *Let f be as in Definition 1, let $A = \partial_x - \frac{1}{2}f$ and $H = -\partial_x^2 + \frac{1}{4}f^2 + \frac{1}{2}f'$. Then $Lv = -\mathrm{e}^{-F/2}H(\mathrm{e}^{F/2}v)$ for every $v \in C^2(\mathbb{R})$, and $H = A^\dagger A$ on $C_c^\infty(\mathbb{R})$. Moreover $Hu_1 = Hu_2 = 0$ and $\mathrm{Wr}(u_2, u_1) = 1$, where $u_1 = \mathrm{e}^{F/2}$ and $u_2(x) = \mathrm{e}^{F(x)/2}\int_x^\infty \mathrm{e}^{-F}$ whenever the latter integral converges.*

Proof Both identities are direct computations, the first by expanding $H(\mathrm{e}^{F/2}v)$ and using $F' = f$, the second by integrating $\langle A^\dagger Au, u\rangle$ by parts. We have $Au_1 = 0$, hence $Hu_1 = 0$; and $u_2 = u_1\int_x^\infty u_1^{-2}$ gives $Hu_2 = 0$ and the Wronskian by reduction of order. □

So an explicit zero-energy solution is available for every drift, which is the structural reason exact answers can be hoped for. We now state the hypotheses under which the passage time is well behaved. They say that the left boundary cannot be reached, that the right one is reached, and that the operator has a discrete spectrum with a summable inverse.

Definition 3 Say that f satisfies (H) if $f \in C^1(\mathbb{R})$, $\int_{-\infty}^0 \mathrm{e}^{-F} = +\infty$, $\int_0^{+\infty} \mathrm{e}^{-F} < +\infty$, and the closure of $A^\dagger A$ on $C_c^\infty(\mathbb{R})$, still written H, is a positive operator with compact resolvent whose inverse is of trace class. Its eigenvalues, in increasing order, are written θ_k, $k \geq 1$, and $T = \sum_k \theta_k^{-1}$.

Under (H) the determinant below is well defined and entire, by Lidskii's theorem [37], and it will turn out to be the reciprocal of a Laplace transform.

Definition 4 For $s \in \mathbb{C}$, let $D(s) = \det(1 + sH^{-1}) = \prod_{k\geq 1}(1 + s/\theta_k)$, and write $D(s) = \sum_{n\geq 0} d_n s^n$.

Proposition 5 is the general form of the answer, and it is the analogue, for an explosion, of a theorem of Kent, to which we owe the mechanism. For a target b interior to the state space, Kent's Corollary 5.1 [29] gives, as the starting point recedes to a left boundary which is not natural, the Laplace transform $\prod_k (1 + s/\lambda_{k,b})^{-1}$, where the $\lambda_{k,b}$ are the eigenvalues of the problem with a Dirichlet condition at b; the hypotheses there bear on the left boundary alone, the summability $\sum_k \lambda_{k,b}^{-1} < \infty$ being [29, Thm. 4.1(e)]. Here the target is the right boundary itself, since the diffusion explodes, so that the statement needs in addition the limit $b \uparrow +\infty$, in which the Dirichlet eigenvalues decrease to those of H on the whole line; that limit is not covered by the hypotheses above. The proof below carries out both limits at once, which is why we give it, and it is also the one place where the role of the explicit zero-energy solution is visible. Its proof uses the solution of $Lv = sv$ normalised at $-\infty$, and the fact that this normalisation is exact at $s = 0$, where $v \equiv 1$ because L annihilates the constants; probabilistically, this is the absence of killing.

Proposition 5 *Let f satisfy* (H). *As $x \to -\infty$, τ_x converges in law to $\sum_{k\geq 1} E_k/\theta_k$, where the E_k are independent exponential variables of mean* 1, *and $\mathbb{E}[\tau_x] \to T$. Moreover $T = \iint_{x<x'} \mathrm{e}^{F(x)-F(x')}\,\mathrm{d}x\,\mathrm{d}x'$.*

Proof The argument is Kent's in substance, on the whole line and with the explosion in place of an interior target. For $s \geq 0$, let φ, χ be real solutions of $(H + s)u = 0$, in L^2 near $-\infty$ and $+\infty$ respectively, with $\mathrm{Wr}(\chi, \varphi) = 1$; then $(H + s)^{-1}$ has the continuous kernel $\varphi(x \wedge y)\chi(x \vee y)$, which is a positive operator, and Mercer's theorem [37, Ch. 3] on a compact interval K gives $\mathrm{Tr}(\mathbf{1}_K(H + s)^{-1}\mathbf{1}_K) = \int_K \varphi\chi$; letting $K \uparrow \mathbb{R}$,

$$\sum_{k\geq 1} \frac{1}{\theta_k + s} = \int_{\mathbb{R}} \varphi\chi. \tag{1}$$

By Lemma 2 and (H), u_1 is in L^2 at $-\infty$ and u_2 at $+\infty$, so (1) at $s = 0$ with $\varphi = u_1$, $\chi = u_2$ gives the last formula for T.

Let $\mathcal{K}g(x) = \int_{-\infty}^{x} \mathrm{e}^{-F(y)} \int_{-\infty}^{y} \mathrm{e}^{F(r)} g(r)\,\mathrm{d}r\,\mathrm{d}y$, let $\Phi = \mathcal{K}1$, which increases to T, and let $u = T - \Phi$, so that $Lu = -1$ and $0 \leq u \leq T$. Let σ_a be the hitting time of a. The scale function $S = \int_0^{\cdot} \mathrm{e}^{-F}$ has $S(-\infty) = -\infty$ and $S(+\infty) < \infty$ by (H) [26, §5.5], so $\mathbb{P}(\sigma_{-m} < \sigma_n) \to 0$ as $m \to \infty$, and Itô's formula at $\sigma_n \wedge \sigma_{-m}$, with u bounded, gives $\mathbb{E}[\sigma_n] = u(x) - u(n)$ for $n > x$. Thus $\sigma_n \uparrow \tau_x$, the explosion is at $+\infty$, and $\mathbb{E}[\tau_x] = u(x) \to T$ as $x \to -\infty$.

Let $s > 0$ and let $v = \sum_{n\geq 0} s^n \mathcal{K}^n 1$, which converges since $|\mathcal{K}^n g| \leq \sup|g|\,\Phi^n/n!$ by induction, and is bounded by e^{sT}. Then v is the solution of $Lv = sv$ with $v \to 1$ at $-\infty$ and $\mathrm{e}^F v' \to 0$ there. Write $V(s) = \lim_{x\to+\infty} v(x)$, which exists and is finite, the two integrals in $\mathcal{K}$ being convergent. Differentiating $Lv = sv$ in s and integrating from the two ends as

in (1) gives $\partial_s \log V(s) = \sum_k (\theta_k + s)^{-1}$; and $V(0) = 1$, since $v \equiv 1$ at $s = 0$. Hence $V = D$. Stopping the bounded martingale $\mathrm{e}^{-st} v(X_t^x)$ at σ_n gives, as $n \to \infty$, $v(x) = D(s)\,\mathbb{E}[\mathrm{e}^{-s\tau_x}]$. As $v(x) \to 1$ when $x \to -\infty$, $\mathbb{E}[\mathrm{e}^{-s\tau_x}] \to \prod_k (1 + s/\theta_k)^{-1}$, the Laplace transform of the claimed sum, which gives convergence in law [17, XIII.1]. □

Two consequences follow at once, and both hold for every drift satisfying (H). The first reads the coefficients of the determinant; the second gives all the cumulants of the passage time.

Proposition 6 *Let f satisfy* (H). *Then $d_0 = 1$ and, for $n \geq 1$,*

$$d_n = \int_{r_n < y_n < r_{n-1} < y_{n-1} < \cdots < r_1 < y_1} \prod_{j=1}^{n} \mathrm{e}^{F(r_j) - F(y_j)}\, \mathrm{d}r_j\, \mathrm{d}y_j. \tag{2}$$

Proof By the proof of Proposition 5, $D(s) = V(s) = \sum_n s^n \lim_{x \to +\infty} \mathcal{K}^n 1(x)$, and iterating the definition of $\mathcal{K}$ gives (2). □

The ordered integrals of Proposition 6 have an exact finite counterpart, which we record because it is the place where the two expansions of the determinant, the one at the origin and the one at infinity, are visible at once; Section 10 returns to this.

Definition 7 For $N \geq 2$ and $x_i, y_i > 0$, let $t_{2i} = (y_i x_{i+1})^{-1}$ and $t_{2i+1} = (x_{i+1} y_{i+1})^{-1}$, let $(\mathsf{A}\psi)(i) = \sqrt{t_{2i}}\,\psi(i) - \sqrt{t_{2i+1}}\,\psi(i+1)$ for $1 \leq i \leq N-1$, with $\psi(N) = 0$, and let $\mathsf{H} = \mathsf{A}^{\mathsf{T}}\mathsf{A}$.

Proposition 8 *With the notation of Definition 7,*

$$\det(1 + s\mathsf{H}^{-1}) = \sum_{n \geq 0} s^n \sum_{i_1 < j_1 \leq i_2 < j_2 \leq \cdots \leq i_n < j_n} y_{i_1} x_{j_1} \cdots y_{i_n} x_{j_n},$$

the inner sum being over the indices in $\{1, \ldots, N\}$ satisfying these inequalities.

Proof The matrix A is upper bidiagonal with positive diagonal, hence invertible, and computing A^{-1} gives $(\mathsf{A}^{-1})_{ij} = \sqrt{y_i x_{j+1}}$ for $i \leq j$, and 0 otherwise. Write $\mathsf{M} = \mathsf{A}^{-1}$, so that $\mathsf{H}^{-1} = \mathsf{M}\mathsf{M}^{\mathsf{T}}$ and $\det(1 + s\mathsf{H}^{-1}) = \sum_n s^n \sum_{|I|=n} \det\big((\mathsf{M}\mathsf{M}^{\mathsf{T}})_{I,I}\big)$, the inner sum being over the n-element subsets of $\{1, \ldots, N-1\}$. By the Cauchy–Binet formula, $\det((\mathsf{M}\mathsf{M}^{\mathsf{T}})_{I,I}) = \sum_{|K|=n} \det(\mathsf{M}_{I,K})^2$. As $\mathsf{M}_{ik} = \sqrt{y_i}\sqrt{x_{k+1}}\,\mathbf{1}_{i \leq k}$, one has $\mathsf{M}_{I,K} = \mathrm{diag}(\sqrt{y_i})_{i \in I}\ \mathsf{U}_{I,K}\ \mathrm{diag}(\sqrt{x_{k+1}})_{k \in K}$, with $\mathsf{U}_{ik} = \mathbf{1}_{i \leq k}$, so that $\det(\mathsf{M}_{I,K})^2 = \prod_{i \in I} y_i \prod_{k \in K} x_{k+1}\, \det(\mathsf{U}_{I,K})^2$. For $I = \{i_1 < \cdots < i_n\}$ and $K = \{k_1 < \cdots < k_n\}$, the 0–1 matrix $\mathsf{U}_{I,K}$ has staircase form, and its determinant is 1 when $k_{a-1} < i_a \leq k_a$ for every a, with $k_0 = 0$, and 0 otherwise. Setting $j_a = k_a + 1$ turns these conditions into $i_1 < j_1 \leq i_2 < j_2 \leq \cdots \leq i_n < j_n$, which gives the sum. □

The second consequence is stated in terms of the spectral zeta function of H, which we therefore introduce; Remark 19 returns to it.

Definition 9 Let f satisfy (H). The spectral zeta function of H is $\zeta(\sigma) = \sum_{k\geq 1} \theta_k^{-\sigma}$, for $\Re\sigma \geq 1$, where the series converges by (H). For the drift of the normal form of Section 3, with its parameter z, it is written ζ_z.

Corollary 10 *Let f satisfy* (H) *and let τ have the limit law of Proposition 5. For every $n \geq 1$, the nth cumulant of τ is $(n-1)!\,\zeta(n)$, and $\zeta(n) = (-1)^{n-1} n\,[s^n] \log D(s)$.*

Proof An exponential variable of rate θ has nth cumulant $(n-1)!\,\theta^{-n}$, and cumulants add over independent summands, which gives the first formula; the second follows by expanding $\log D(s) = \sum_k \log(1 + s/\theta_k)$ at the origin. □

In particular $\mathbb{E}[\tau] = \zeta(1) = T$ and $\mathrm{Var}(\tau) = \zeta(2)$. All of this is general. What the next sections add is that, for the normal form, these quantities can be evaluated, and the reason is already visible: a quadratic drift has a cubic primitive, which makes the double integral above collapse, and gives the operator a quartic potential, whose six Stokes sectors are what Section 9 will use.

3 The normal form and its oscillator

We now take the drift of the saddle-node normal form, with its bifurcation parameter, and record that it satisfies the hypotheses of Section 2. The operator produced by Lemma 2 is then a quartic oscillator, and it is the object computed with in the rest of the paper. We allow a complex parameter, which costs nothing and is needed for Theorem 16.

Definition 11 For $z, x \in \mathbb{C}$, set $f_z(x) = x^2 - z$ and $F_z(x) = x^3/3 - zx$, and set $V_z = \frac{1}{4} f_z^2 + \frac{1}{2} f_z'$, that is, $V_z(x) = \frac{1}{4}(x^2 - z)^2 + x$.

Definition 12 For $z \in \mathbb{C}$, H_z is the operator $-\mathrm{d}^2/\mathrm{d}x^2 + V_z$ on $L^2(\mathbb{R})$, with domain that of the closure of H_0 on $C_c^\infty(\mathbb{R})$. Its eigenvalues, counted with algebraic multiplicity, are written $\theta_k(z)$; for real z they are taken in increasing order. For $(s, z) \in \mathbb{C}^2$, let $D(s, z) = \det(1 + sH_z^{-1})$, and write $D(s, z) = \sum_{n\geq 0} d_n(z) s^n$.

By Definition 11 and Lemma 2, H_z is the operator attached to the drift f_z, and it is a supersymmetric quartic oscillator with superpotential $f_z/2$, passing from a single well to a double well as z grows [10]. The z-dependent lower-order terms are relatively H_0-bounded with bound 0: expanding $\|H_0 u\|^2$ on $C_c^\infty(\mathbb{R})$ gives $\|x^4 u\| \leq C(\|H_0 u\| + \|u\|)$, and $x^2 \leq \epsilon x^4 + C_\epsilon$ [33, Ch. X]; hence the domain does not depend on z, the family is holomorphic of type (A) [11, Ch. VII], and H_z has compact resolvent since H_0 does. For real z, H_z is self-adjoint with simple eigenvalues.

Lemma 13 records that this operator satisfies the hypotheses of Section 2, so that everything proved there applies to it.

Lemma 13 *For every $z \in \mathbb{C}$, H_z is invertible, and $z \mapsto H_z^{-1}$ is holomorphic from $\mathbb{C}$ into the trace-class operators. For real z, $\theta_k(z) > 0$ for all k, and f_z satisfies* (H) *of Definition 3.*

Proof By Lemma 2 and $\Re F_z(x) \sim x^3/3$, $u_1 = \mathrm{e}^{F_z/2}$ is in L^2 only at $-\infty$ and u_2 only at $+\infty$. A zero mode would be proportional to both, contradicting $\mathrm{Wr}(u_2, u_1) = 1$; as the resolvent is compact, H_z is invertible. By the Weyl law $\theta_k(0) \asymp k^{4/3}$ [39], H_0^{-1} is of trace class, and so is $H_z^{-1} = H_0^{-1}(H_z H_0^{-1})^{-1}$, the domains being equal. Near z_0, $H_z^{-1} = H_{z_0}^{-1}(1 + E(z))^{-1}$, with $E(z) = \big(-\frac{1}{2}(z - z_0)x^2 + \frac{1}{4}(z^2 - z_0^2)\big)H_{z_0}^{-1}$ bounded, holomorphic and $E(z_0) = 0$. For real z and $u \in C_c^\infty(\mathbb{R})$, $\langle H_z u, u\rangle = \int |u' - \frac{1}{2} f_z u|^2 \geq 0$, so $\theta_k(z) > 0$ by invertibility. Finally, $-F_z(x) \to +\infty$ as $x \to -\infty$ and $-F_z(x) \to -\infty$ as $x \to +\infty$, with $|F_z(x)| \sim |x|^3/3$, so the two integrals of (H) diverge and converge respectively. □

So Section 2 applies verbatim to the normal form, and we restate its conclusion in this case before evaluating the quantities it involves.

4 The exact law

Theorem 15 is Proposition 5 for the normal form. We state it separately because the rates are now the eigenvalues of an explicit oscillator, on which the rest of the paper computes.

Definition 14 Let $z \in \mathbb{R}$ and let W be a standard Brownian motion. For $x \in \mathbb{R}$, let X^x be the solution of $\mathrm{d}X_t = (X_t^2 - z)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t$ started at x, up to its explosion time τ_x.

Theorem 15 *Let $z \in \mathbb{R}$. As $x \to -\infty$, τ_x converges in law to $\sum_{k\geq 1} E_k/\theta_k(z)$, where the E_k are independent exponential variables of mean 1, and $\mathbb{E}[\tau_x] \to \sum_k 1/\theta_k(z)$. The Laplace transform of the limit law is $1/D(\cdot, z)$.*

Proof By Lemma 13, f_z satisfies (H), and Proposition 5 applies, the determinant of Definition 4 being $D(\cdot, z)$ by Definition 12. □

Figure 2(a) compares the law with a simulation. What this setting adds to the expansion is the operator: its rates are the eigenvalues of a Schrödinger operator which the supersymmetric factorisation makes explicit, and which for the normal form is a quartic oscillator whose spectral quantities the following sections evaluate in closed form. That evaluation, and not the expansion, is what the rest of the paper is about.

5 The mean, in closed form, and why for this drift alone

By Theorem 15 the mean passage time is the trace of H_z^{-1}, which Proposition 5 gives as a double integral. We now evaluate it. The computation rests on two identities. For

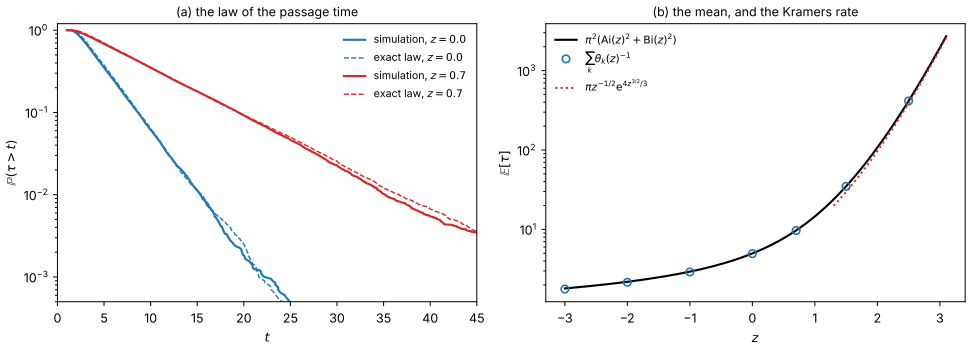


Fig. 2 (a) The survival function of the passage time, at $z = 0$ and $z = 0.7$. Solid: $2 \cdot 10^4$ simulated paths of Definition 14, started at $x = -10$ with step $5 \cdot 10^{-4}$, the explosion being declared at $x = 10$. Dashed: the law of Theorem 15, sampled with the first 500 eigenvalues of H_z, the remaining part of the trace being added as a constant. The tail is exponential, at the rate $\theta_1(z)$. (b) The mean passage time: the closed form of Theorem 16 (solid), the sum of the first 500 computed eigenvalues together with an estimate of the tail of that sum (circles), and the Kramers rate of Corollary 21 (dotted).

$y, \ell \in \mathbb{R}$ and $z \in \mathbb{C}$, a direct expansion gives

$$F_z(y + \ell/2) - F_z(y - \ell/2) = y^2\ell + \frac{\ell^3}{12} - z\ell, \tag{3}$$

and the integral representation of [31, §9.11], extended from real z by analytic continuation, gives for every $z \in \mathbb{C}$

$$\sqrt{\pi} \int_0^\infty \ell^{-1/2}\, \mathrm{e}^{z\ell - \ell^3/12}\, \mathrm{d}\ell = \pi^2 \big(\mathrm{Ai}(z)^2 + \mathrm{Bi}(z)^2\big). \tag{4}$$

Here Ai, Bi are the Airy functions [31, Ch. 9], and $a_1 > a_2 > \dots$ denote the zeros of Ai. We also write $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$.

Theorem 16 *For every $z \in \mathbb{C}$,*

$$\sum_{k \geq 1} \frac{1}{\theta_k(z)} = \pi^2 \big(\mathrm{Ai}(z)^2 + \mathrm{Bi}(z)^2\big) = 4\pi^2\, \mathrm{Ai}\big(\omega z\big)\mathrm{Ai}\big(\omega^{-1} z\big).$$

This function of z vanishes exactly at the simple zeros $|a_k| \mathrm{e}^{\pm \mathrm{i}\pi/3}$, $k \geq 1$.

Proof For real z, put $x = y - \ell/2$, $x' = y + \ell/2$ in the double integral of Proposition 5. By (3) and a Gaussian integration in y, it becomes the left-hand side of (4). By Lemma 13, $\mathrm{Tr}\, H_z^{-1}$ is entire, so the identity extends to $\mathbb{C}$, and $\mathrm{Tr}\, H_z^{-1} = \sum_k \theta_k(z)^{-1}$ by Lidskii's theorem [37]. The connection formula $\mathrm{Ai}(\mathrm{e}^{\mp 2\pi\mathrm{i}/3} z) = \frac{1}{2}\mathrm{e}^{\mp \mathrm{i}\pi/3}(\mathrm{Ai}(z) \pm \mathrm{i}\mathrm{Bi}(z))$ [31, §9.2] gives the factorisation. All zeros of Ai are real, negative and simple [31, §9.9], so the zeros of the factors are $a_k \mathrm{e}^{\mp 2\pi\mathrm{i}/3} = |a_k| \mathrm{e}^{\pm \mathrm{i}\pi/3}$. □

The trace will be used throughout, and we give it a name.

Definition 17 For $z \in \mathbb{C}$, let $T(z) = \sum_k \theta_k(z)^{-1}$.

For real z, Theorems 15 and 16 together give the mean passage time in closed form, and the extension to complex z is what makes the factorisation above visible. This closed form appears, for real z, in [24, Eq. (15)]; what is added here is that it is the spectral trace of H_z, that it extends to complex z, and that it is the first cumulant of the exact law of Theorem 15.

The single step of the proof which is not general is the Gaussian integration in y, and Proposition 18 shows that it is available for the quadratic drift alone. We state it for polynomial drifts, which is the case where the question has a clean answer. It decides the fate of this route and of no other: it says that the mechanism which produces the closed form is available for the quadratic drift alone, not that no closed form exists for another drift by other means. The case $d = 1$ is covered by the statement and is empty, an affine drift never satisfying (H). The substitution $x = y - \ell/2$, $x' = y + \ell/2$ turns the double integral of Proposition 5 into the one below, with G the increment of the primitive.

Proposition 18 *Let f be a polynomial of degree $d \geq 1$, let F be a primitive of f, and let $G(y, \ell) = F(y + \ell/2) - F(y - \ell/2)$. In the double integral $\iint_{\ell>0} \mathrm{e}^{-G(y,\ell)}\, \mathrm{d}y\, \mathrm{d}\ell$, the inner integration, in y, is Gaussian if and only if $d = 2$.*

Proof Write $F(u) = \sum_{j \leq d+1} a_j u^j$ with $a_{d+1} \neq 0$, and $h = \ell/2$. In $G = \sum_j a_j[(y+h)^j - (y-h)^j]$, the binomial terms even in h cancel, so that $G = 2\sum_j a_j \sum_{i \text{ odd}} \binom{j}{i} y^{j-i} h^i$. The highest power of y is y^d, with coefficient $2a_{d+1}(d+1)h \neq 0$ for $\ell \neq 0$. The exponent of the inner integrand is a polynomial of degree d in y, which is quadratic exactly when $d = 2$. □

So, for a polynomial drift, the substitution $x = y - \ell/2$, $x' = y + \ell/2$ reduces the double integral of Proposition 5 to a Gaussian integration in y, followed by a single integral in ℓ, exactly when the drift is quadratic, that is, exactly for the saddle-node normal form. For any other degree the inner integration is not Gaussian, and this route produces no closed form; we make no claim about other routes. The universal normal form and the one case the method solves therefore coincide, which is why the two have always appeared together in the literature.

6 The cumulants

By Corollary 10, all the cumulants of the passage time are the values of the spectral zeta function at the positive integers, and by Proposition 6 they are read off the ordered integrals of $\mathrm{e}^{\pm F_z}$. For the normal form the first of them is Theorem 16, so that $\mathbb{E}[\tau] = \pi^2(\mathrm{Ai}(z)^2 + \mathrm{Bi}(z)^2)$ and $\mathrm{Var}(\tau) = \zeta_z(2) = \sum_k \theta_k(z)^{-2}$. At the symmetric point $z = 0$ the variance is also in closed form: Corollary 37 below derives it from the functional equation of Section 9, giving $\mathrm{Var}(\tau) = \mathbb{E}[\tau]^2/3$ with $\mathbb{E}[\tau] = 3^{-1/3}\Gamma(1/3)^2$, so that the coefficient of variation is $1/\sqrt{3}$.

Remark 19 The zeta function of H_z is also governed at the other end. By the Mellin argument of Voros [41, 42], the Mellin transform of $\log D(\cdot, z)$ is $\pi\zeta_z(\sigma)/(\sigma \sin \pi\sigma)$ up to an entire factor, so that the values of ζ_z at the positive integers come from the Taylor coefficients of D at the origin, which are the ordered integrals of Proposition 6, and its poles, at $\sigma_m = 3/4 - m/2$, from the coefficients of the expansion at infinity of Proposition 24; in particular $\zeta_z(-n) = 0$ for integers $n \geq 0$, and the zeta-regularised determinant $\exp(-\zeta_z'(0))$ of [42] equals 1, which is the normalisation $D(0, z) = 1$.

7 The tail

The law of Theorem 15 has an exponential tail whose rate is the smallest eigenvalue, since $\mathbb{P}(\tau > t) \geq \mathbb{P}(E_1 > \theta_1(z)t)$ and the remaining factors are summable. In the double-well regime $z \to +\infty$ this rate is exponentially small, and it is the Kramers rate [30, Eq. 3.10], the inverse of the exponentially small energy of Witten [43]. Proposition 20 bounds the correction, and Corollary 21 gives the rate with its prefactor; the closed form of Theorem 16 is what makes the prefactor accessible.

Proposition 20 *As $z \to +\infty$, $T(z) - 1/\theta_1(z) = O(\ln z/\sqrt{z})$.*

Proof Let $\phi_0 = \mathrm{e}^{F_z/2}\mathbf{1}_{x<\sqrt{z}}$ and $I(t) = \int_{-\infty}^{t} \mathrm{e}^{F_z}$. The kernel $u_1(x\wedge y)u_2(x\vee y)$ and the variational principle give $1/\theta_1 \geq \langle\phi_0, H_z^{-1}\phi_0\rangle/\|\phi_0\|^2 = I(\sqrt{z})^{-1}\int_{\mathbb{R}} \mathrm{e}^{-F_z(t)} I(t\wedge\sqrt{z})^2 \mathrm{d}t$. Since $T = \int_{\mathbb{R}} \mathrm{e}^{-F_z} I$ by Proposition 5, one obtains $T - 1/\theta_1 \leq 2\mathcal{I}_1 + \mathcal{I}_2$, with $\mathcal{I}_1 = \iint_{\sqrt{z}<r<t} \mathrm{e}^{F_z(r)-F_z(t)}$ and $\mathcal{I}_2 = \iint_{-\sqrt{z}<t<r<\sqrt{z}} \mathrm{e}^{F_z(r)-F_z(t)}$, the region $t < -\sqrt{z}$ giving $\mathcal{I}_1$ again because F_z is odd. Use (3) with $\ell = |t-r|$. In $\mathcal{I}_1$, let $p = (t+r)/2 - \sqrt{z} > \ell/2$; then $y^2 - z \geq \max(p^2, 2\sqrt{z}p)$, and the integral in p is at most $\min(\frac{1}{2}\sqrt{\pi/\ell}, \mathrm{e}^{-\sqrt{z}\ell^2}/(2\sqrt{z}\ell))$. In $\mathcal{I}_2$, let $p = \sqrt{z} - |y| \geq \ell/2$; then $z - y^2 \geq \sqrt{z}p$ and $\ell^2/12 \leq (z-y^2)/3$, so that the exponent is at most $-\frac{2}{3}\sqrt{z}\ell p$. Cutting at $\ell = 1/z$ gives $O(\ln z/\sqrt{z})$ for both. □

This gives the exponentially small ground energy, with its prefactor.

Corollary 21 *As $z \to +\infty$, $\theta_1(z) \sim \pi^{-1}\sqrt{z}\,\mathrm{e}^{-\frac{4}{3}z^{3/2}}$.*

Proof As $T(z) \to \infty$, Proposition 20 gives $\theta_1(z)T(z) \to 1$, and Theorem 16 with $\mathrm{Bi}(z)^2 \sim \mathrm{e}^{\frac{4}{3}z^{3/2}}/(\pi\sqrt{z})$ [31, §9.7] concludes. □

The exponent is twice the instanton action $\frac{2}{3}z^{3/2}$: this is the instanton–anti-instanton effect of supersymmetric quantum mechanics [43], here with its prefactor. Numerically, $\theta_1(5)T(5) - 1 = 5.3 \cdot 10^{-7}$.

8 Small times

The tail just described is the behaviour of τ at large times, and it came from the smallest eigenvalue. The behaviour at small times comes from the whole spectrum at once, and it is read at the other end of the Laplace transform. The cumulants of Section 6 read that transform at the origin $s = 0$, where it describes the bulk of the law. The opposite end $s \to \infty$ of the same transform governs the behaviour of that law near $t = 0$, that is, the probability of an unusually quick passage, and this section turns one into the other. At that end the determinant is exponentially large, and its leading behaviour, which is all that the statement about τ requires, costs nothing but the Weyl law. The statement is therefore independent of the full expansion, which is recalled afterwards because Sections 6 and 9 need it.

Lemma 22 *Let $z \in \mathbb{R}$ and let $N_z(\theta) = \#\{k : \theta_k(z) \le \theta\}$. Then, as $\theta \to +\infty$,*

$$N_z(\theta) \sim \kappa\, \theta^{3/4}, \qquad \kappa = \frac{\mathrm{B}(\frac14, \frac32)}{\pi\sqrt{2}},$$

and, as $s \to +\infty$ along the real axis,

$$\log D(s, z) \sim c_0\, s^{3/4}, \qquad c_0 = \pi\sqrt{2}\,\kappa = \mathrm{B}\big(\tfrac14, \tfrac32\big) = \frac{\sqrt{2}\,\Gamma(\frac14)^2}{3\sqrt{\pi}} = 3.49607\ldots.$$

Proof The potential V_z of Definition 11 is a quartic tending to $+\infty$, so the Weyl law in the form of [39] gives $N_z(\theta) = \pi^{-1} \int_{V_z<\theta} (\theta - V_z)^{1/2} + o(\theta^{3/4})$. As $V_z - x^4/4$ has degree 2, replacing V_z by $x^4/4$ changes that integral by $o(\theta^{3/4})$, and the substitution $x = (4\theta)^{1/4} t$ turns it into $\pi^{-1} 4^{1/4} \theta^{3/4} \int_{-1}^{1} (1 - t^4)^{1/2} \mathrm{d}t$. Setting $t^4 = v$ gives $\int_0^1 (1 - t^4)^{1/2} \mathrm{d}t = \frac14 \mathrm{B}(\frac14, \frac32)$, whence the value of κ, with $4^{1/4} = \sqrt{2}$.

For the second claim, $-\partial_\theta \log(1 + s/\theta) = s/(\theta(\theta + s))$, and N_z vanishes below θ_1 while $N_z(\theta) \log(1 + s/\theta) \to 0$, so an integration by parts followed by $\theta = su$ gives

$$\log D(s, z) = \int_0^\infty \log\Big(1 + \frac{s}{\theta}\Big) \mathrm{d}N_z(\theta) = \int_0^\infty \frac{N_z(\theta)\, s\, \mathrm{d}\theta}{\theta(\theta + s)} = \int_0^\infty \frac{N_z(su)}{u(1 + u)}\, \mathrm{d}u.$$

By the first claim, $N_z(\theta) \le C_z \theta^{3/4}$ on $(0, \infty)$, and $\int_0^\infty u^{-1/4}(1 + u)^{-1} \mathrm{d}u = \pi/\sin(3\pi/4) = \pi\sqrt{2}$, so dominated convergence applies to $s^{-3/4} N_z(su) \to \kappa u^{3/4}$ and gives $s^{-3/4} \log D(s, z) \to \pi\sqrt{2}\,\kappa$. Finally $\mathrm{B}(\frac14, \frac32) = \Gamma(\frac14)\Gamma(\frac32)/\Gamma(\frac74)$, and $\Gamma(\frac32) = \frac12\sqrt{\pi}$, $\Gamma(\frac74) = \frac34\Gamma(\frac34)$, $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$ give the closed form. □

Why the leading coefficient should be a Beta value is visible in the proof: it is the area under the quartic turning-point profile $(1 - t^4)^{1/2}$. We now recall the full expansion, which refines this and which the rotations of Section 9 require. It is governed by the Riccati equation of $(H_z + s)\psi = 0$. Set $Q = \frac14 f_z^2 + x + s$.

Definition 23 Let $S_0 = Q^{1/2}$, with the principal branch, for the s such that $Q(\mathbb{R}) \cap (-\infty, 0] = \emptyset$, and, for $n \ge 1$, $2S_0 S_n = -S_{n-1}' - \sum_{i=1}^{n-1} S_i S_{n-i}$. Let $P = \sum_{n \ge 0} S_{2n}$, as a formal series.

The series $S = \sum_n S_n$ solves $S^2 + S' = Q$ formally, and its even-indexed part P gives the formal solutions $P^{-1/2}\exp(\pm\int P)$. For a decaying potential, the terms of P at large s are the densities of the KdV hierarchy [20]. Here the potential grows, the integrals of these densities diverge, and one must rescale $x = (4s)^{1/4}y$. Proposition 24 gives the resulting expansion.

Proposition 24 *Let $\delta > 0$. As $s \to \infty$ with $|\arg s| \le \pi - \delta$, uniformly for z in compact sets of $\mathbb{C}$, for every $N \ge 0$, with principal powers,*

$$D(s,z) = \exp\Big(\sum_{m=0}^{N} c_m(z)\, s^{3/4-m/2} + O\big(s^{1/4-N/2}\big)\Big),$$

where c_m is a polynomial in z, and

$$c_0 = \mathrm{B}\big(\tfrac14, \tfrac32\big), \quad c_1 = \frac{\sqrt2\,\Gamma(\frac34)^2}{\sqrt\pi}\, z, \quad c_2 = \frac{\sqrt2\,\Gamma(\frac14)^2}{32\sqrt\pi}\, z^2, \quad c_3 = \frac{\sqrt2\,\Gamma(\frac34)^2}{32\sqrt\pi}\,(z^3 - 4).$$

Proof This is the standard WKB expansion of the quartic equation, by the formal series of Definition 23 and a comparison argument on the Wronskian. The companion paper [27] proves it in three steps, for a general linear term ν in the potential, the coefficient c_3 reading there $(z^3 + 4 - 8\nu^2)$; the statement above is the case $\nu = 1$. □

The coefficient c_0 is the one of Lemma 22, reached there from the Weyl law and here from the Riccati equation: the agreement of the two routes on the value $\mathrm{B}(\frac14, \frac32)$ is a check on both. Half-integer values of m being absent, the exponent $3/4 - m/2$ never vanishes, so the expansion has no constant term: D is the determinant normalised at infinity in the sense of [42]. The singular values of H_z^{-1} are $O(k^{-4/3})$, as those of H_0^{-1}, so $\log|D(s,z)| \le C_z(1 + |s|^{3/4})$ on $\mathbb{C}$ [37]; with $c_0 > 0$, $D(\cdot, z)$ has order exactly $3/4$.

Proposition 25 turns this end into a statement about τ, which completes the description of its law: Theorem 16 and Corollary 10 describe it in the bulk, Corollary 21 in the tail, and the next statement near the origin.

Proposition 25 *Let $z \in \mathbb{R}$ and let τ have the limit law of Theorem 15. As $t \downarrow 0$,*

$$-\log \mathbb{P}(\tau \le t) \sim \frac{A}{t^3}, \qquad A = \frac{27}{256}c_0^4 = \frac{\Gamma(1/4)^8}{192\pi^2} = 15.7560\ldots,$$

and A does not depend on z.

Proof By Theorem 15, $\mathbb{E}[\mathrm{e}^{-s\tau}] = 1/D(s,z)$, so that $-\log\mathbb{E}[\mathrm{e}^{-s\tau}] = \log D(s,z)$, and Lemma 22 gives $\log D(s,z) \sim c_0 s^{3/4}$ as $s \to +\infty$ along the real axis. The law of τ is carried by $[0,\infty)$, so Kasahara's exponential Tauberian theorem [2, Thm. 4.12.9] applies: with $\alpha = 3$, so that $\alpha/(\alpha+1) = 3/4$, the equivalence above is equivalent to $-\log\mathbb{P}(\tau \le t) \sim At^{-\alpha}$, where $c_0 = (\alpha+1)\alpha^{-\alpha/(\alpha+1)}A^{1/(\alpha+1)} = 4\cdot 3^{-3/4}A^{1/4}$. Solving for A gives $A = 27c_0^4/256$, and the value of c_0 gives the closed form. □

The duality behind this is that of Legendre: by the Chernoff bound, $-\log \mathbb{P}(\tau \leq t) \geq \sup_{s>0}(\log D(s,z) - st)$, and replacing $\log D$ by $c_0 s^{3/4}$ makes the supremum $27c_0^4/(256t^3)$, attained at $s = (3c_0/4t)^4$; the Tauberian theorem is what makes this bound sharp to leading order. That A does not depend on z has a direct reading: a passage quick enough leaves no time to feel the bifurcation parameter, which enters only the subsequent terms of the expansion. For $z = 0$ and $s = 5$, the truncation of Proposition 24 at $m = 6$ gives 11.64637 for $\log D = 11.64629$.

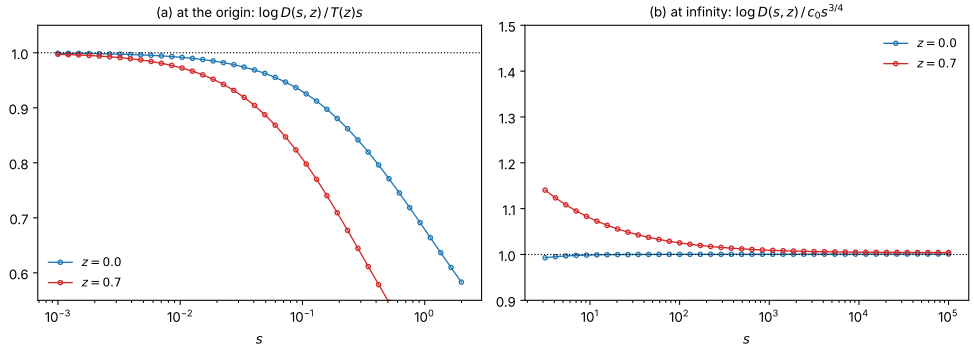


Fig. 3 The two ends of $\log D(s,z)$, at $z = 0$ and $z = 0.7$, computed from the first 800 eigenvalues together with the Weyl tail of the series. (a) Divided by $T(z)s$, it tends to 1 as $s \to 0$: the origin is governed by the trace, Theorem 16. (b) Divided by $c_0 s^{3/4}$, it tends to 1 as $s \to \infty$, and to the same limit for both values of the parameter, the dependence on z sitting in the subsequent coefficients of Proposition 24; this is what makes the constant of Proposition 25 independent of z.

Figure 3 shows the two ends. The description of the law is now complete: the bulk by Theorem 16 and Corollary 10, the tail by Corollary 21, the small times by Proposition 25.

9 A functional equation, and where it comes from

The Laplace transform of the passage time is $1/D(\cdot, z)$, and this function satisfies an exact algebraic relation, of a different nature from the preceding results: it does not come from the diffusion but from the degree of the potential. That a Laplace transform of a hitting time should satisfy a cubic algebraic identity is not something the probabilistic side of the problem would suggest, and the identity has a consequence there: it yields the variance of the passage time in closed form at the symmetric point, hence the coefficient of variation $1/\sqrt{3}$ (Corollary 37), a value the literature had reached only by a direct computation [35, 36], [30, Eq. 3.5]. Rotating x by a cube root of unity permutes three copies of the problem, and the three resulting determinants are linked by one cubic equation. We prove it for a quartic oscillator with an arbitrary linear term, which is the generality in which the mechanism is visible, and then specialise. A companion paper [27] studies that general family for its own sake, and several of the statements below are proved there; we recall the arguments which are short.

Definition 26 For $\nu \in \mathbb{R}$ and $z \in \mathbb{C}$, let $H_{z,\nu} = -\mathrm{d}^2/\mathrm{d}x^2 + \frac{1}{4}(x^2 - z)^2 + \nu x$, with the domain of H_z, let $\mu = \mathrm{e}^{\mathrm{i}\pi\nu}$, and let $\eta_k = (-1)^{k+1}$ for $k \in \mathbb{Z}$.

For ψ entire, consider

$$\psi'' = \big(\tfrac{1}{4}(x^2 - z)^2 + \nu x + s\big)\psi. \tag{5}$$

For $\nu = 1$, this is $(H_z + s)\psi = 0$. Up to scaling and translation, every real quartic with positive leading coefficient is of the form $\frac{1}{4}(x^2 - z)^2 + \nu x + s$ with real (ν, z, s).

9.1 The Stokes solutions: a technical part

This subsection is technical. The cubic relation reduces to one fact: the Wronskians of six solutions of (5) form a 3×3 matrix of rank at most two. Here we construct these solutions and identify the entries of the matrix with the determinant. On a first reading, one can retain the statements and skip the proofs; the main ones are the three relations (7), (8), (9) and Proposition 31.

We use the solutions of (5) which are subdominant in six sectors, provided by the next statement.

Proposition 27 *For each $k \in \mathbb{Z}$, there is a solution $y_k = y_k(\cdot; s, z)$ of (5), entire in (x, s, z) and analytic in ν, such that for every $\delta > 0$,*

$$y_k(x) = x^{\eta_k\nu - 1}\mathrm{e}^{\eta_k F_z(x)/2}(1 + o(1)), \quad y_k'(x) = \tfrac{1}{2}\eta_k f_z(x) x^{\eta_k\nu - 1}\mathrm{e}^{\eta_k F_z(x)/2}(1 + o(1)), \tag{6}$$

as $x \to \infty$ with $|\arg x - k\pi/3| \le \pi/2 - \delta$, where $\arg x$ is taken in this interval, on the universal cover, uniformly for (s, z) in compact sets. This property determines y_k.

Proof The argument is Sibuya's; it is also in [27]. Setting $x = 2^{1/3}\mathrm{e}^{\mathrm{i}k\pi/3}t$ turns (5) into $\varphi'' = P_k(t)\varphi$, with P_k monic of degree 4 and polynomial in (s, z), whose coefficients are $b_1 = 0$, $b_2 = -2^{-2/3}\mathrm{e}^{4\mathrm{i}k\pi/3}z$ and $b_3 = (-1)^k\nu$ in the notation of [34, Ch. 2], so that the exponent $-1 - b_3$ equals $\eta_k\nu - 1$. The subdominant solution $Y_0(t; P_k)$ of [34, Ch. 2, Thm. 6.1] then gives $y_k(x) = 2^{(\eta_k\nu-1)/3}\mathrm{e}^{\mathrm{i}k\pi(\eta_k\nu-1)/3}\,Y_0(t; P_k)$, which is determined by its decay on $\arg x = k\pi/3$, where a second independent solution grows. □

By (6) and the uniqueness, three relations hold. In the overlap of consecutive sectors,

$$\mathrm{Wr}(y_k, y_{k+1}) = (-1)^k. \tag{7}$$

Comparing the determinations of $x^{\eta_k\nu-1}$ on $\arg x = k\pi/3$ and $\arg x = k\pi/3 + 2\pi$,

$$y_{k+6} = \mu^{2\eta_k} y_k. \tag{8}$$

If ψ solves (5) at $(\omega^2 s, \omega z)$, then $\psi(\omega^{-1}\cdot)$ solves it at (s, z), since $\omega^3 = 1$, and $F_{\omega z}(\omega^{-1}x) = F_z(x)$. Hence

$$y_{k+2}(x; s, z) = \mathrm{e}^{2\mathrm{i}\pi(\eta_k\nu-1)/3}\, y_k(\omega^{-1}x; \omega^2 s, \omega z). \tag{9}$$

The solutions decaying on the two ends of the real axis are y_0 and, up to a phase, y_3. Their Wronskian is the spectral function of $H_{z,\nu}$.

Definition 28 Let $\psi_- = y_0$, $\psi_+ = -\mu^{-1} y_3$ and $\mathcal{E}_\nu(s,z) = \mathrm{Wr}(\psi_-, \psi_+)$.

Lemma 29 collects the properties of these objects.

Lemma 29 *Let $\nu \in \mathbb{R}$ and $(s,z) \in \mathbb{C}^2$.*

(i) As $x \to +\infty$, $\psi_-(x) \sim x^{-\nu-1} \mathrm{e}^{-F_z(x)/2}$, and ψ_- is, up to a factor, the only solution of (5) in L^2 at $+\infty$.
(ii) As $x \to -\infty$, $\psi_+(x) \sim |x|^{\nu-1} \mathrm{e}^{F_z(x)/2}$, and ψ_+ is, up to a factor, the only solution of (5) in L^2 at $-\infty$.
(iii) $\mathcal{E}_\nu$ is entire, and it vanishes exactly when $-s$ is an eigenvalue of $H_{z,\nu}$.
(iv) $\mathcal{E}_{-\nu} = \mathcal{E}_\nu$.

Proof The arguments are short, and are also in [27]. Items (i) and (ii) follow from (6), a second solution growing at each end, and (iii) from Proposition 27 together with these items. For (iv), the reflection $x \mapsto -x$ exchanges the two solutions of the parameters ν and $-\nu$, since F_z is odd. □

We now compute the s-dependence of $\mathcal{E}_\nu$ and identify $\mathcal{E}_1$. The identification rests on the behaviour of $\psi_\pm$ at the opposite end, which Lemma 30 describes.

Lemma 30 *Let $\nu \in \mathbb{R}$ and $(s,z) \in \mathbb{C}^2$. As $x \to +\infty$, $\psi_+ = \mathcal{E}_\nu y_1 + O(x^{-\nu-1}\mathrm{e}^{-F_z/2})$ and $\psi_+' = \mathcal{E}_\nu y_1' + O(x^{1-\nu}\mathrm{e}^{-F_z/2})$. As $x \to -\infty$, $\psi_- = O(|x|^{-\nu-1}\mathrm{e}^{-F_z/2})$ and $\psi_-' = O(|x|^{1-\nu}\mathrm{e}^{-F_z/2})$.*

Proof Write $\psi_+ = a y_0 + b y_1$; then (7) gives $b = \mathrm{Wr}(y_0, \psi_+) = \mathcal{E}_\nu$, and (6) gives the first claim, $\mathbb{R}_+$ lying in the sectors of y_0 and y_1. The second claim follows in the same way from $y_0 = a' y_2 + b' y_3$, the negative axis lying in the sectors of y_2 and y_3. □

Proposition 31 *(i) For all $(s,z) \in \mathbb{C}^2$, $\mathcal{E}_1(s,z) = D(s,z)$.*
(ii) For $\nu \in \mathbb{R}$, $z \in \mathbb{R}$ and $s, s_0 > -\theta_1$, with θ_k the eigenvalues of $H_{z,\nu}$ in increasing order, $\mathcal{E}_\nu(s,z)/\mathcal{E}_\nu(s_0,z) = \prod_k (\theta_k + s)/(\theta_k + s_0)$.

Proof Let $\nu \in \mathbb{R}$, $z \in \mathbb{R}$, $s > -\theta_1$, and let a dot denote ∂_s. Then $\mathcal{E}_\nu \neq 0$ by Lemma 29(iii). Differentiating (5) in s, $\mathrm{Wr}(\dot\psi_-, \psi_+)' = -\psi_-\psi_+ = -\mathrm{Wr}(\psi_-, \dot\psi_+)'$. As the asymptotics of Proposition 27 are uniform in s, Cauchy's estimates give that $\dot\psi_\pm$, $\dot\psi_\pm'$ are o of the leading terms of $\psi_\pm$, $\psi_\pm'$ at $\mp\infty$ in (6). So, by Lemma 30 and (6), the first Wronskian tends to 0 at $+\infty$ and the second at $-\infty$; their sum is $\partial_s \mathcal{E}_\nu$, so integrating from the two ends gives $\partial_s \mathcal{E}_\nu = \int_{\mathbb{R}} \psi_-\psi_+$. For real s, $\psi_\pm$ are real by uniqueness, and (1), for the positive operator

$H_{z,\nu}+s$, with $\varphi=\psi_+$ and $\chi=\psi_-/\mathcal{E}_\nu$, gives $\partial_s \ln \mathcal{E}_\nu = \sum_k (\theta_k+s)^{-1}$, which is (ii). For $\nu=1$, $\theta_1>0$ by Lemma 13, and at $s=0$, uniqueness and Lemma 2 give $\psi_+=u_1$, $\psi_-=u_2$, so $\mathcal{E}_1(0,z)=1=D(0,z)$. Hence $\mathcal{E}_1=D$ on $[0,\infty)\times\mathbb{R}$, and on $\mathbb{C}^2$ by the identity theorem. □

This ends the technical part.

9.2 The cubic relation

The rank argument now takes a few lines.

Theorem 32 *For all $\nu\in\mathbb{R}$ and $(s,z)\in\mathbb{C}^2$, with $\mathcal{E}^{(0)}=\mathcal{E}_\nu(s,z)$, $\mathcal{E}^{(1)}=\mathcal{E}_\nu(\omega^2 s,\omega z)$, $\mathcal{E}^{(2)}=\mathcal{E}_\nu(\omega s,\omega^2 z)$,*

$$\mathcal{E}^{(0)}\mathcal{E}^{(1)}\mathcal{E}^{(2)}-\mathcal{E}^{(0)}-\mathcal{E}^{(1)}-\mathcal{E}^{(2)}=2\cos\pi\nu.$$

Proof The argument is short and the present section rests on it, so we give it here; it is also in [27]. By Definition 28, $\mathrm{Wr}(y_0,y_3)=-\mu\mathcal{E}_\nu$. Applying (9) to y_0 and y_3, whose phases multiply to $\mathrm{e}^{-4\mathrm{i}\pi/3}=\omega$ because $\eta_0=-\eta_3$, and using that a Wronskian in x is ω^{-1} times the Wronskian in $\omega^{-1}x$, gives $\mathrm{Wr}(y_2,y_5)(s,z)=\mathrm{Wr}(y_0,y_3)(\omega^2 s,\omega z)$, and then $\mathrm{Wr}(y_4,y_7)(s,z)=\mathrm{Wr}(y_2,y_5)(\omega^2 s,\omega z)$. With (7) and (8) for the remaining entries, the matrix of $\mathrm{Wr}(u,v)$, for $u\in\{y_0,y_2,y_4\}$ and $v\in\{y_3,y_5,y_7\}$, is

$$\begin{pmatrix} -\mu\mathcal{E}^{(0)} & \mu^2 & \mu^2 \\ 1 & -\mu\mathcal{E}^{(1)} & \mu^2 \\ 1 & 1 & -\mu\mathcal{E}^{(2)} \end{pmatrix}.$$

Its rows are linearly dependent, the solutions of (5) forming a space of dimension two, and its determinant is $-\mu^3\big(\mathcal{E}^{(0)}\mathcal{E}^{(1)}\mathcal{E}^{(2)}-\mathcal{E}^{(0)}-\mathcal{E}^{(1)}-\mathcal{E}^{(2)}-\mu-\mu^{-1}\big)$. □

What is special at $\nu=1$ is that the normalisation is known for every z, so that the general relation becomes an equation on the determinant itself: this is the equation announced in the introduction.

Theorem 33 *For all $(s,z)\in\mathbb{C}^2$, with $D_0=D(s,z)$, $D_1=D(\omega^2 s,\omega z)$, $D_2=D(\omega s,\omega^2 z)$,*

$$D_0D_1D_2=D_0+D_1+D_2-2.$$

Proof Take $\nu=1$ in Theorem 32, where $2\cos\pi\nu=-2$, and use $\mathcal{E}_1=D$, which is Proposition 31(i). □

Read on its first coefficients, the equation reproduces a classical identity of the Airy functions and extends it; Remark 36 does this.

The conjugation of Lemma 2 gives the cubic relation a direct meaning, which Remark 34 records.

Remark 34 By Lemma 2, $(H_z + s)\psi = 0$ is equivalent to $L_z v = sv$ with $L_z = \partial_x^2 + f_z \partial_x$ and $v = \mathrm{e}^{-F_z/2}\psi$. For $\nu = 1$, $\mu = -1$, and the Wronskians $\mathrm{Wr}(u, v)$, $u \in \{y_0, y_2, y_4\}$, $v \in \{y_3, y_5, y_7\}$, form the matrix $\left(\begin{smallmatrix} D_0 & 1 & 1 \\ 1 & D_1 & 1 \\ 1 & 1 & D_2 \end{smallmatrix}\right)$, of rank at most two. It has a direct meaning: along the rays $\arg x = 0, 2\pi/3, 4\pi/3$, every solution v has a limit, namely $\mathrm{Wr}(y_0, \psi)$, $\mathrm{Wr}(y_2, \psi)$, $\mathrm{Wr}(y_4, \psi)$, by (6) and (7). The matrix lists these three limits for the three solutions $\psi = y_3, y_5, y_7$.

At the symmetric point $z = 0$, the rotation acts on s alone, and the relation becomes a statement on one spectrum, for every ν. Its first coefficients give Proposition 35.

Proposition 35 *For every $\nu \in \mathbb{R}$, $\mathcal{E}_\nu(0, 0) = 2\cos(\pi\nu/3)$. The number 0 is an eigenvalue of $H_{0,\nu}$ exactly for $\nu \in \frac{3}{2} + 3\mathbb{Z}$. For the other ν, let $c = 2\cos(\pi\nu/3)$, let θ_k be the eigenvalues of $H_{0,\nu}$ and $q_m = \sum_k \theta_k^{-m}$ for $m \geq 1$. Then*

$$(2c^2 - 2)\, q_3 = q_1^3 - 3q_1 q_2.$$

Proof Let $E(s) = \mathcal{E}_\nu(s, 0)$. At $s = 0$, the three arguments coincide, so $c_\nu = E(0)$ satisfies $c_\nu^3 - 3c_\nu = 2\cos\pi\nu$ by Theorem 32, that is, c_ν is one of the $2\cos(\pi(\nu+2k)/3)$. As $\mathcal{E}_\nu$ is analytic in ν (Proposition 27), one value of k holds for all ν, and as $c_{-\nu} = c_\nu$ by Lemma 29(iv), $k = 0$. The zeros of E follow from Lemma 29(iii). If $c \neq 0$, then Proposition 31(ii), with $s_0 > -\theta_1$, shows that E is a constant times the entire function $\prod_k (1 + s/\theta_k)$, and evaluating at $s = 0$ gives $E(s) = c\sum_n e_n s^n$, where e_n is the nth elementary symmetric function of the θ_k^{-1}. The coefficient of s^3 in Theorem 32 gives $c^3 q_3 = 3ce_3$, and Newton's identity $6e_3 = q_1^3 - 3q_1 q_2 + 2q_3$ concludes. □

Comparison with the literature.

The mechanism is known, and what follows is an extension. The constant Wronskians (7), evaluated at $x = 0$, give the bilinear identity of Voros [42, Eq. (43)], between half-line determinants, for adjacent rotations. Theorem 32 is a further step, of Plücker type: it uses third neighbours, on the full line. For the homogeneous potential, that is, $z = 0$ and $\nu = 0$, it is the closed relation of Kamata [25, Eq. (44b)], obtained by fusion from the A_3 T-system, and Proposition 35 at $\nu = 0$ is his sum rule $q_1^3 - 3q_1 q_2 - 6q_3 = 0$. The Y-system of the quartic potential [23] underlies the general case; see also [8, 38] for functional relations and supersymmetric factorisations in other families. What Theorem 32 and Proposition 35 add is the closed form on the full line for the whole family (z, ν), for a single function, with the constant $2\cos\pi\nu$.

Remark 36 The coefficient of s in Theorem 33 is an identity. The coefficient of s^2 reads $\omega T(z)T(\omega^2 z) + \omega^2 T(z)T(\omega z) + T(\omega^2 z)T(\omega z) = 0$. With $\mathrm{A}_j = \mathrm{Ai}(\omega^j z)$, Theorem 16 gives $T(z) = 4\pi^2 \mathrm{A}_1\mathrm{A}_2$, $T(\omega^2 z) = 4\pi^2 \mathrm{A}_0\mathrm{A}_1$, $T(\omega z) = 4\pi^2 \mathrm{A}_0\mathrm{A}_2$, and the relation becomes $\mathrm{A}_0\mathrm{A}_1\mathrm{A}_2(\mathrm{A}_0 + \omega\mathrm{A}_1 + \omega^2\mathrm{A}_2) = 0$, the classical connection formula [31, §9.2]. The functional equation thus extends this relation to all orders in s.

At $\nu = 1$, $c = 1$ in Proposition 35: the term in q_3 disappears, and the trace gives the value of q_1. The sum rule then takes a closed form, which is due to $z = 0$ and to the cubic relation, and supersymmetry only contributes $c = 1$ and the value of q_1.

Corollary 37 *Let $\theta_k = \theta_k(0)$, $d_n = d_n(0)$ and $q_m = \sum_k \theta_k^{-m}$. Then*

$$\prod_{k\geq 1}\Big(1+\frac{s^3}{\theta_k^3}\Big) = 1 + 3\sum_{n\geq 1} d_{3n}s^{3n}, \qquad q_1 = 3^{-1/3}\Gamma(1/3)^2, \qquad q_2 = \frac{q_1^2}{3} = 3^{-5/3}\Gamma(1/3)^4.$$

Proof At $z = 0$, $D_0D_1D_2 = \prod_k(1+s^3/\theta_k^3)$ and $D_0+D_1+D_2 = 3\sum_{3|n} d_n s^n$, so Theorem 33 gives the first formula. Proposition 35 at $\nu = 1$, where $c = 1$, gives $q_1^3 = 3q_1q_2$, with $q_1 > 0$. The value of q_1 is Theorem 16 with $\mathrm{Ai}(0)^2 + \mathrm{Bi}(0)^2 = 4\mathrm{Ai}(0)^2$. □

9.3 The monodromy surface

Theorem 32 says that the triple of rotated determinants lies on a cubic surface, and that surface is not an arbitrary one.

Remark 38 For $\nu \in \mathbb{R}$, let $\mathscr{S}_\nu$ be the surface $X_0X_1X_2 - X_0 - X_1 - X_2 = 2\cos\pi\nu$ in $\mathbb{C}^3$, and let $\mathbf{E}_\nu(s,z) = (\mathcal{E}^{(0)}, \mathcal{E}^{(1)}, \mathcal{E}^{(2)})$, so that Theorem 32 reads $\mathbf{E}_\nu(s,z) \in \mathscr{S}_\nu$. This surface is the monodromy manifold of the second Painlevé equation $u'' = 2u^3 + xu - \alpha$, with $\alpha = \nu - \frac{1}{2}$, in the normalisation of van der Put and Saito [16], and its coordinates are, up to signs, the Stokes multipliers of the Flaschka–Newell pair [12, 18] read at a pole of a solution; the companion paper [27] proves this identification, together with the fact that an eigenvalue of $H_{z,\nu}$ corresponds to a vanishing multiplier, and that the level of an eigenvalue is the rank of the corresponding pole. The surface is smooth for $\nu \notin \mathbb{Z}$ and has one singular point for $\nu \in \mathbb{Z}$, where the monodromy becomes reducible; the singular fibres of this family are versal deformations of an A_1 singularity [4, App. B], and the monodromy of the family has been computed by Roffelsen and Stokes [15]. The symplectic nature of the map $\mathbf{E}_\nu$, for ν outside the integers, follows from the general theory of the irregular Riemann–Hilbert map [13, Cor. 1], in the form established for the Painlevé II hierarchy by Bertola and Tarricone [14, Thm. 2.5], with the constant $2\pi\mathrm{i}$. What this identification buys is the name of the surface, the reading of the constant $2\cos\pi\nu$ as its parameter, and the open problems of Section 10; the functional equation itself is used in Corollary 37.

At integer ν the surface is singular, and the determinant reaches that singular point. Proposition 40, which we use in Section 9.4 to describe the curve traced by the determinant, says where; it is stated in terms of the quasi-exactly solvable points, which we define first.

Definition 39 For $n \geq 0$ and $z \in \mathbb{C}$, let $\mathcal{T}_{n,z}v = v'' + f_z v' - nxv$. A number s is a quasi-exactly solvable point of level n if it is an eigenvalue of $\mathcal{T}_{n,z}$ on polynomials of degree at most n.

For $\nu = n+1$ the substitution $\psi = \mathrm{e}^{F_z/2}v$ turns (5) into $\mathcal{T}_{n,z}v = sv$, and the left-hand side preserves polynomials of degree at most n; this is the quasi-exactly solvable structure [1, 40]. For $n = 0$ the only such point is $s = 0$, for every z, which is the supersymmetric case.

Proposition 40 *Let $n \geq 0$ and $z \in \mathbb{C}$. If s is a quasi-exactly solvable point of level n, then $\mathcal{E}_{n+1}(s,z) = (-1)^n$ and $\mathbf{E}_{n+1}(s,z) = (-1)^n(1,1,1)$, the singular point of $\mathscr{S}_{n+1}$. In particular $\mathbf{E}_1(0,z) = (1,1,1)$ for every z, and in the coordinates $\zeta_j = X_j - 1$, $\mathscr{S}_1$ is the surface $\zeta_0\zeta_1 + \zeta_1\zeta_2 + \zeta_2\zeta_0 + \zeta_0\zeta_1\zeta_2 = 0$.*

Proof The term of highest degree of $\mathcal{T}_{n,z}x^m$ is $(m-n)x^{m+1}$, so every eigenvector has degree exactly n. For a monic eigenvector v, $\psi = \mathrm{e}^{F_z/2}v$ solves (5) with $\nu = n+1$, and $\psi = (-1)^n\psi_+$ by uniqueness; letting $x \to +\infty$ in $\mathrm{Wr}(\psi_-, \psi)$ gives $\mathcal{E}_{n+1} = (-1)^n$. If v is such an eigenvector, then $\psi(\omega\,\cdot) = \mathrm{e}^{F_{\omega z}/2}v(\omega\,\cdot)$ solves (5) at $(\omega^2 s, \omega z)$, so $\omega^2 s$ is a quasi-exactly solvable point of level n for ωz, and likewise ωs for $\omega^2 z$, which gives the triple. The last claim is the case $n = 0$, $s = 0$, with an expansion. □

So, at $\nu = 1$, $\mathbf{E}_1$ sends the whole line $s = 0$ to the singular point of $\mathscr{S}_1$, and this is what the supersymmetric factorisation contributes on the surface: a point known for every value of the bifurcation parameter.

The rotations of Theorem 33 turn the leading exponent of Proposition 24 into an imaginary one, which gives a product formula.

Corollary 41 *Let $z \in \mathbb{C}$. As $s \to +\infty$, with D_1, D_2 as in Theorem 33, $D_1D_2 = 1 + O\bigl(\exp(-\frac{1}{2}c_0 s^{3/4})\bigr)$.*

Proof Theorem 33 gives $D_1D_2 - 1 = (D_1 + D_2 - 2)/D_0$. By Proposition 24 at (s,z), $(\omega^2 s, \omega z)$ and $(\omega s, \omega^2 z)$, where $\arg = 0, \mp 2\pi/3$, $\log|D_0| \sim c_0 s^{3/4}$, and for D_1, D_2 the leading term of the exponent is $c_0(\mathrm{e}^{\mp 2\mathrm{i}\pi/3}s)^{3/4} = \mp \mathrm{i}c_0 s^{3/4}$, which is imaginary, so that $\log|D_j| = O(s^{1/4})$. □

9.4 The two ends on the surface

Both ends being known, we can place them on the Stokes surface. By Propositions 40 and 24 and Corollary 41, the curve $s \mapsto \mathbf{E}_1(s,z)$, $s \geq 0$, leaves the singular point of $\mathscr{S}_1$ at $s = 0$ and goes to infinity along the region $X_0 \to \infty$, $X_1X_2 \to 1$. More is true of the product of Corollary 41 than the exponentially small bound: the coefficients computed in Proposition 24 consist of monomials z^a with $2a + m \equiv 0 \pmod 3$, which makes the corresponding terms of $\log D_1$ and $\log D_2$ cancel in pairs. We have not proved that this holds for every m. For $s < 0$, the relation also gives one exact statement on the way: at $s = -\theta_k(z)$, $D_0 = 0$, so that $D_1 + D_2 = 2$, and for real z the curve meets the line $\Re X_1 = 1$. Figure 4 shows this curve. One Laplace transform is thus drawn as one path on a fixed surface, whose singular point it leaves at $s = 0$ and whose distinguished

circle it joins at infinity, the eigenvalues of the oscillator being the parameters at which it crosses a line; the bifurcation parameter moves the path and not the surface.

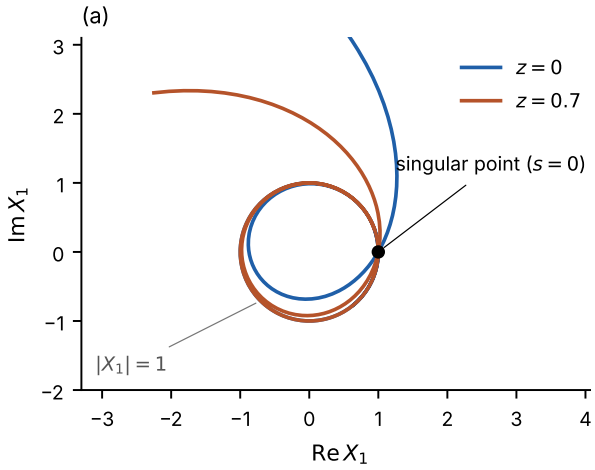


Fig. 4 The curve $s \mapsto X_1 = D(\omega^2 s, \omega z)$ for $-0.75 \leq s \leq 6$, at $z = 0$ and $z = 0.7$. It leaves the singular point of $\mathscr{S}_1$ at $s = 0$; for $s > 0$ it joins the circle $|X_1| = 1$, in accordance with Corollary 41, as $X_2 = \overline{X_1}$ for real z; for $s < 0$ it leaves the circle, and it meets the line $\Re X_1 = 1$ at $s = -\theta_1(z)$.

10 Conclusion and open problems

The question of the title has an answer in two parts, which are the first and the third of the levels announced in the introduction. A passage time of this kind is computable through the spectrum whenever the drift sends the particle in at one end and out at the other, in the sense of (H): its law is then a sum of independent exponential variables whose rates are the eigenvalues of a Schrödinger operator, and its cumulants are the values of the corresponding spectral zeta function. That expansion follows Kent, for a target reached by explosion rather than an interior one; what makes it usable is that the operator is supersymmetric, the drift being its superpotential, so that a zero-energy solution is explicit for every drift. By the route followed here it is computable in closed form, among polynomial drifts, for the quadratic one alone, which is the normal form of the saddle-node bifurcation; what decides this is that a quadratic drift has a cubic primitive, and nothing else.

For that normal form the law is described at both ends and in the bulk: the mean is $\pi^2(\mathrm{Ai}(z)^2 + \mathrm{Bi}(z)^2)$, a formula which holds for every complex value of the parameter and in which the factorisation into Airy functions of rotated arguments then becomes visible, the tail decays at the Kramers rate with its prefactor, and the probability of an unusually quick passage decays like $\exp(-\Gamma(1/4)^8/(192\pi^2 t^3))$, with a constant

which does not depend on the bifurcation parameter. The fourth fact announced in the introduction is of another kind: the Laplace transform satisfies an exact cubic functional equation, which gives the variance in closed form at the symmetric point and places the transform on the monodromy surface of the second Painlevé equation. The particle of the introduction, entering from the far left and lingering where the equilibrium has vanished, thus leaves after a time whose law is known exactly for any drift that lets it in and out, and known in closed form because its drift is the one the bifurcation imposes. Several questions remain.

1. *Higher sums.* Find a closed form, if any, for $\mathrm{Var}(\tau) = \zeta_z(2)$ when $z \neq 0$; the coefficient of s^3 in Theorem 33 gives one linear relation between the three rotated values.
2. *Other drifts.* For a polynomial drift of degree d, the potential has degree $2d$ and the equation has $2d+2$ Stokes sectors, so that the Wronskian matrix of Remark 34 becomes $(d+1)\times(d+1)$, still of rank at most two. Determine the resulting system of relations between the rotated determinants, and whether it places them on the Stokes manifold of the corresponding member of the Painlevé II hierarchy [14]. Proposition 18 shows that no closed form for the mean is to be expected there by the same route; what replaces it is open.
3. *Solving the functional equation.* Decide whether Theorem 33, with suitable analyticity and growth conditions, determines D, as in the thermodynamic Bethe ansatz [9], and whether it admits a nonlinear integral form. The resurgent structure in z should also be understood [42].
4. *Quantum cubic relation.* The surface of Remark 38 is quantised by the algebra of the second Painlevé equation of [32, Eq. (6.38)], see also [4], with relations $\mathcal{X}_0\mathcal{X}_1 - q\mathcal{X}_1\mathcal{X}_0 = 1-q$ and their cyclic images, in which $\mathcal{X}_0\mathcal{X}_1\mathcal{X}_2 - \mathcal{X}_0 - q\mathcal{X}_1 - \mathcal{X}_2$ is central; the Flaschka–Newell bracket has been identified with a cluster bracket [14]. Find an operator form of Theorem 33 realising this algebra.
5. *Nonlocal and local charges.* The ordered integrals of Proposition 6 are nonlocal in F_z, whereas the terms of the expansion at infinity are local; in integrable systems, this is the opposition between nonlocal and local charges [3]. Proposition 8 is a finite model where both are visible: with $y_i = \hbar\, \mathrm{e}^{F_z(i\hbar)}$ and $x_i = \hbar\, \mathrm{e}^{-F_z(i\hbar)}$, the operator A discretises $-A$ and the sums of that proposition are Riemann sums of (2), while at $s=\infty$ the same polynomial gives $\mathrm{Tr}\,\mathsf{H}^n$, local in the t_j. The open point is the limit: that second expansion holds only for $|s| > \|\mathsf{H}\|$, of order $\hbar^{-2}$, and it does not survive $\hbar \to 0$. Explain the opposition for the oscillator. In the algebra of the lattice sine-Gordon model [21], both families become commuting operators, through a monodromy [22].

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Appendix A Numerical checks

Two methods are used. For complex z, where H_z is complex symmetric, we diagonalise it in a Hermite basis, which is reliable for the first few hundred eigenvalues and is what Table A1 rests on: there $D(s,z)$ is computed from the first 220 eigenvalues, the

omitted tail being expanded to second order, its first moment being known exactly from Theorem 16 and its second taken from Lemma 22. Everywhere else z is real and we use finite differences on a box, which resolve the high modes correctly and give several thousand eigenvalues, those whose classical turning point $(4\theta)^{1/4}$ lies well inside the box being retained. Table A1 gives the relative residual of Theorem 33, which is compatible with that truncation. The coefficients of Proposition 24 were computed symbolically up to $m = 6$ and checked against a direct integration of the Riccati equation for $z = 0$, 0.7, 1.5, and Corollary 41 by a Taylor computation of the Wronskians, for $s \le 40$, where $|D_1 D_2 - 1|$ falls below 10^{-20}. A 45-digit Taylor integration from asymptotic series at $x = \pm 11$ gives residuals at most 10^{-41} at all the points of Table A1.

The cubic relation for the general family, Theorem 32, is checked numerically in [27]. With the method used there, we also checked $\mathcal{E}_2 = -1$ and $\mathcal{E}_3 = 1$ at the quasi-exactly solvable points of levels 1 and 2, up to 10^{-41}.

Theorem 15 was checked against a direct simulation of the diffusion, as described in Figure 2. At $z = 0.7$, with $2 \cdot 10^4$ paths and no censoring, the simulated standard deviation is 7.396 against the exact value $\zeta_z(2)^{1/2} = 7.404$, and the deciles and the 99th percentile agree to a few percent, the residual difference being accounted for by the finite starting point, which lowers the mean from $T(0.7) = 9.7033$ to the simulated 9.568. Theorem 16 was checked in two ways. First against the spectrum: with the 3006 eigenvalues below the box threshold at $z = 0$ and the Weyl tail of Lemma 22, the sum $\sum_k \theta_k^{-1}$ reproduces $\pi^2(\mathrm{Ai}(0)^2 + \mathrm{Bi}(0)^2) = 4.976054$ to 10^{-5} in relative terms; the coarser estimate used for the circles of Figure 2(b), which stops at 500 eigenvalues and fits the tail, is within two percent across that range and improves as z grows. Second, after the Gaussian integration of its proof, against the single integral $\int_0^\infty \sqrt{\pi/\ell}\, \mathrm{e}^{z\ell - \ell^3/12} \mathrm{d}\ell$, which reproduces the closed form to 16 digits for $-3 \le z \le 3.5$. The Weyl constant of Lemma 22 was checked on $[-26, 26]$ with step $6.5 \cdot 10^{-4}$: $N_z(\theta)\theta^{-3/4}$ gives 0.7875 at $z = 0$ and 0.7885 at $z = 0.7$ for $1000 \le k \le 3000$, against $\kappa = 0.786894$, the residual being of the order of the subsequent Weyl term. The Hermite basis gives it independently at low order, $\theta_{100}(0) = 634.7$ and $\theta_{200}(0) = 1604.6$ against $(k/\kappa)^{4/3} = 638.9$ and 1610.0. Proposition 25 was checked at $z = 0$ by computing $\log D(s, 0)$ from the eigenvalues below the box threshold together with the Weyl tail, taken as an integral; at the truncation the computed eigenvalue and the Weyl value $(k/\kappa)^{4/3}$ agree to 0.14%, which is what makes the splice legitimate. Then $\log D(s, 0)/s^{3/4}$ gives 3.49611 at $s = 10^2$ and 3.49608 at $s = 10^8$, against $c_0 = 3.496077$; and $t^3 \sup_{s>0}(\log D(s, 0) - st)$ gives 15.751, 15.757 and 15.756 at $t = 1$, 0.1 and 0.03, against $A = 15.75606$, the maximiser agreeing with the predicted $(3c_0/4t)^4$ at all three. The vanishing of $\zeta_z(-n)$ (Remark 19) was checked on the heat trace at $z = 0$ and $z = 0.7$: $\sum_k \mathrm{e}^{-t\theta_k}$ minus the terms in $t^{-\sigma_m}$, $m \le 3$, tends to 0 as $t \to 0$, with no constant and no linear term. Proposition 8 was checked numerically for $3 \le N \le 7$, with agreement to machine precision, and so was the closed form of A^{-1} used in its proof.

(s,z)	$(1,0)$	$(2,0)$	$(1,0.7)$	$(1,0.5+0.8\mathrm{i})$	$(0.7+0.4\mathrm{i},1.2)$
$\lvert D(s,z)\rvert$	29.31	330.0	73.72	43.48	76.72
relative residual	$2.0\cdot10^{-5}$	$7.0\cdot10^{-5}$	$1.9\cdot10^{-5}$	$1.9\cdot10^{-5}$	$1.2\cdot10^{-5}$

Table A1 Test of the functional equation, with D computed from the truncated spectrum. The independent high-precision integration described above reduces the residual to 10^{-41} at the same five points.

Appendix B Notation

Table B2 lists the main notation, with the place where it is introduced.

$f_z(x)=x^2-z$, $F_z(x)=\frac{1}{3}x^3-zx$	the function of the supersymmetric factorisation and its primitive
H_z, $H_{z,\nu}$	the operators $-\partial_x^2+\frac{1}{4}f_z^2+x$ and $-\partial_x^2+\frac{1}{4}f_z^2+\nu x$
$\theta_k(z)$	the eigenvalues of H_z, or of $H_{z,\nu}$ when ν is fixed
$D(s,z)$, d_n	the spectral determinant $\det(1+sH_z^{-1})$ and its coefficients
$T(z)$	the trace $\sum_k \theta_k(z)^{-1}$
ω, μ, η_k	$\mathrm{e}^{2\pi\mathrm{i}/3}$, $\mathrm{e}^{\mathrm{i}\pi\nu}$, $(-1)^{k+1}$
y_k, $\psi_\pm$	the Stokes solutions, Proposition 27, and the real solutions, Definition 28
$\mathcal{E}_\nu$, $\mathcal{E}^{(j)}$	the Wronskian of the two normalised solutions and its rotations, Definition 28 and Theorem 32; it equals D at $\nu=1$
f, F, L, H, A	an arbitrary drift, a primitive, the generator, the Schrödinger operator and its factor, Definition 1 and Lemma 2
(H)	the boundary and spectral hypotheses, Definition 3
τ_x	the explosion time, Definitions 1 and 14
ζ, ζ_z	the spectral zeta function, Definition 9
$\mathscr{S}_\nu$, $\mathbf{E}_\nu$	the monodromy surface and the map to it, Remark 38
$\mathcal{T}_{n,z}$	the quasi-exactly solvable operator, Definition 39
c_m, c_0	the coefficients at infinity, Proposition 24; the first is also Lemma 22
N_z, κ	the eigenvalue counting function and its Weyl constant, Lemma 22

Table B2 Main notation.

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Data availability. No experimental data were generated. The numerical results can be reproduced from the formulas in the text and from the scripts, which are available from the author on reasonable request.

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